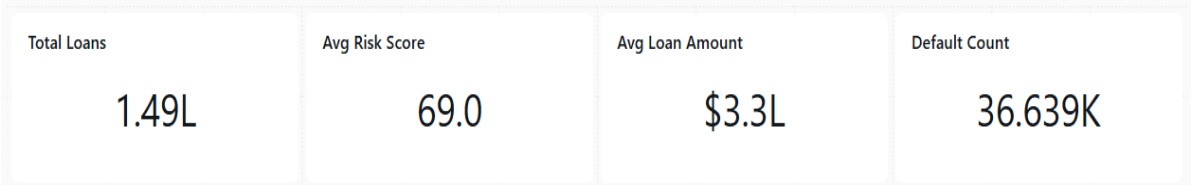
Credit Risk Assessment Pipeline Dashboards

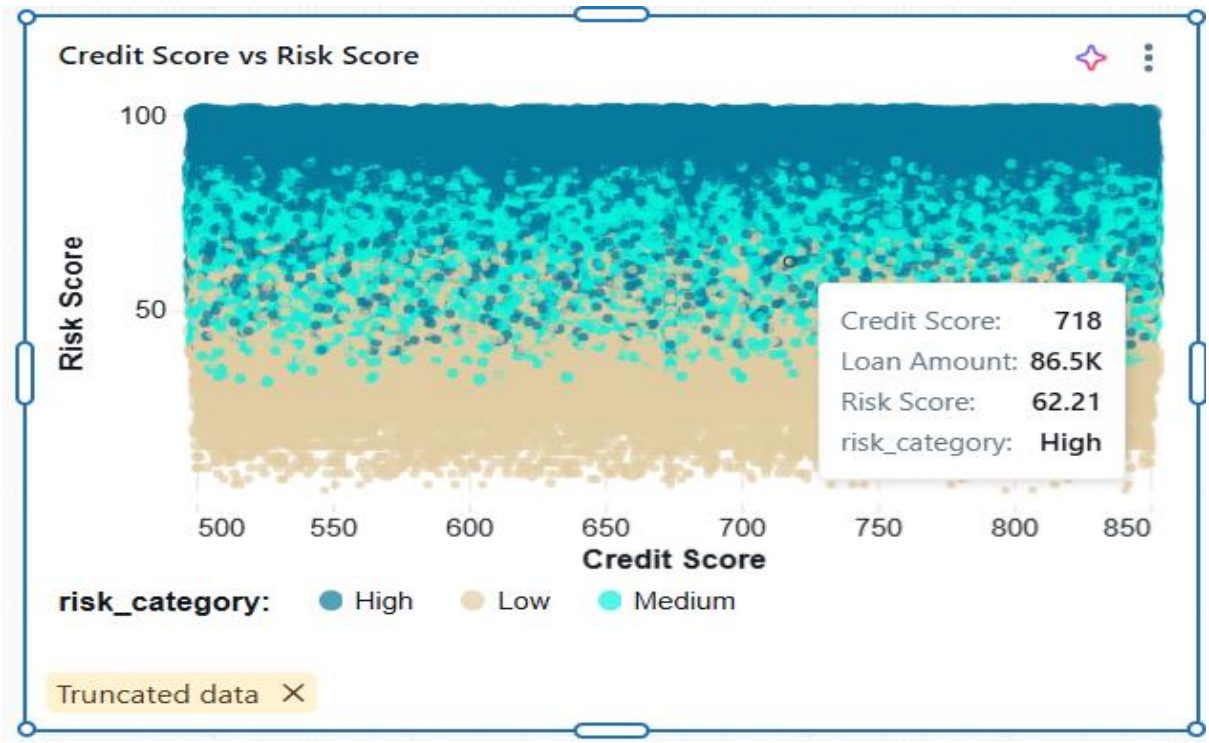
1. KPI Cards

KPI cards display key metrics such as total loans, average risk score, average loan amount, and total defaults. These metrics give a quick overview of the portfolio health. They help stakeholders understand overall trends without deep analysis. For example, a high average risk score may indicate increased exposure. These KPIs are essential for monitoring performance at a glance. They also support faster decision-making in dashboards.



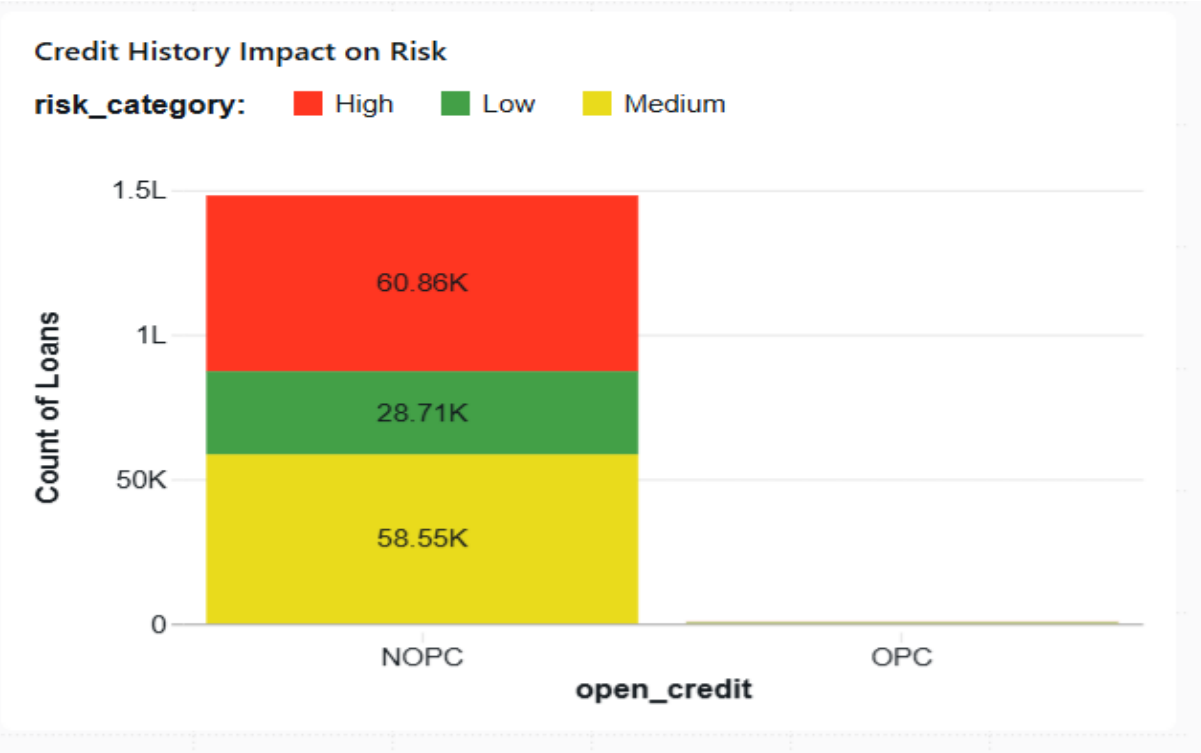
2. Loan Purpose Analysis

This chart analyzes loan distribution across different purposes along with associated risk categories. It helps determine which types of loans are more likely to be risky. For example, personal or emergency loans may show higher risk compared to home loans. Understanding this helps financial institutions tailor risk strategies for each loan purpose. It also assists in adjusting interest rates and approval criteria. This improves overall portfolio quality.



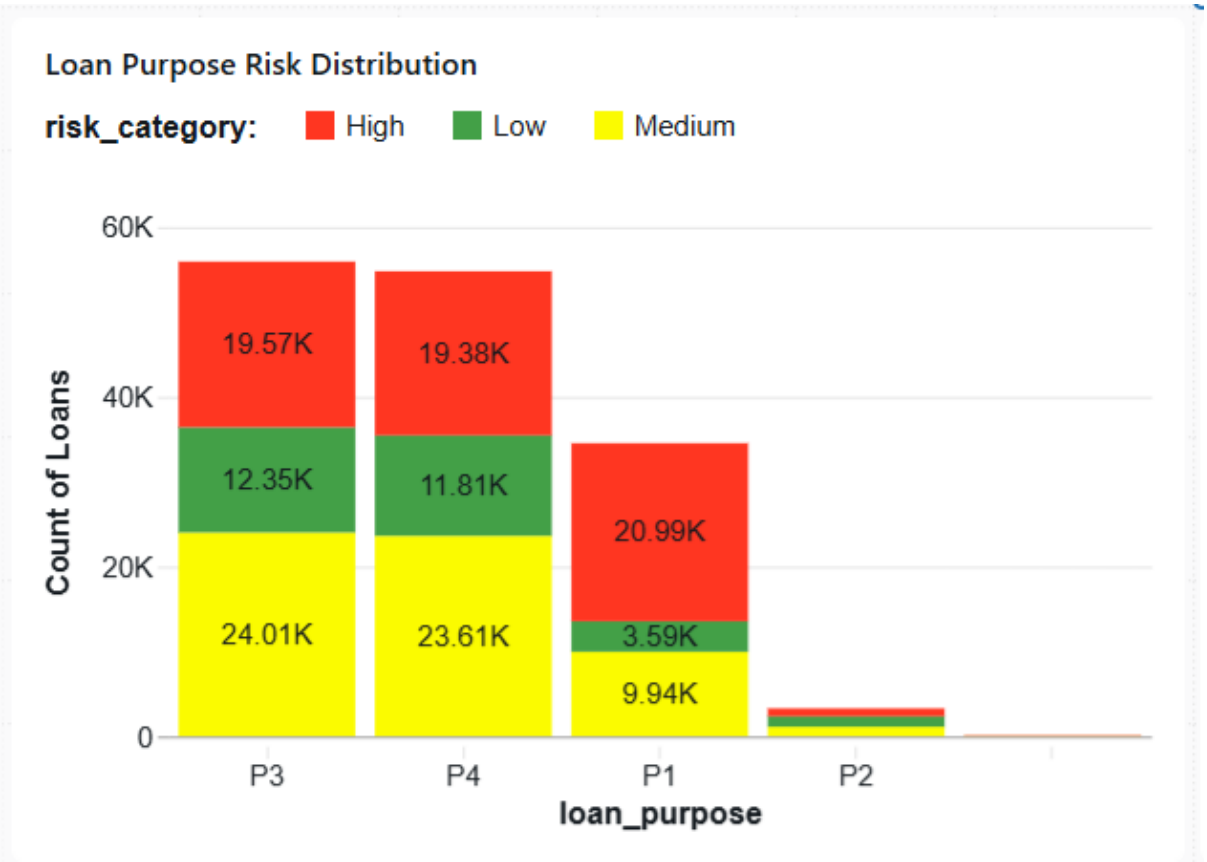
3. Credit History Impact

This visualization uses proxies like open_credit or debt-to-income ratio to analyze credit history impact. It helps understand how borrowing patterns relate to risk levels. A higher number of open credits or high debt ratios may indicate financial stress. By segmenting these factors with risk categories, lenders can detect risky credit behavior. This insight enhances creditworthiness evaluation. It also supports better loan approval decisions and risk control.



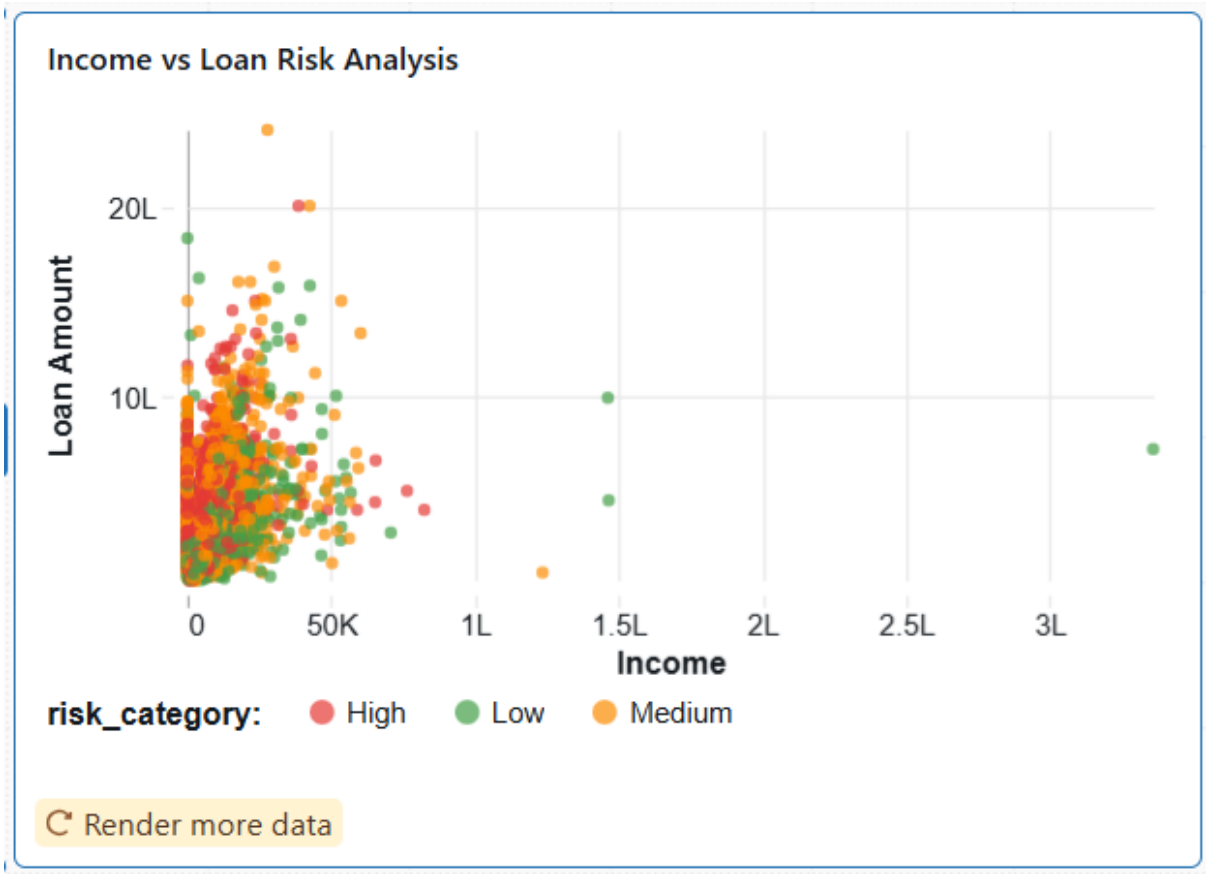
4. **Credit Score Impact**

This scatter plot visualizes how credit scores influence the calculated risk score of borrowers. Typically, lower credit scores are associated with higher risk scores. The plot helps identify patterns, trends, and outliers in borrower behavior. Adding loan amount as a size dimension provides additional context on exposure. This analysis is essential for validating the effectiveness of credit scoring models. It also helps lenders refine credit policies and scoring thresholds.



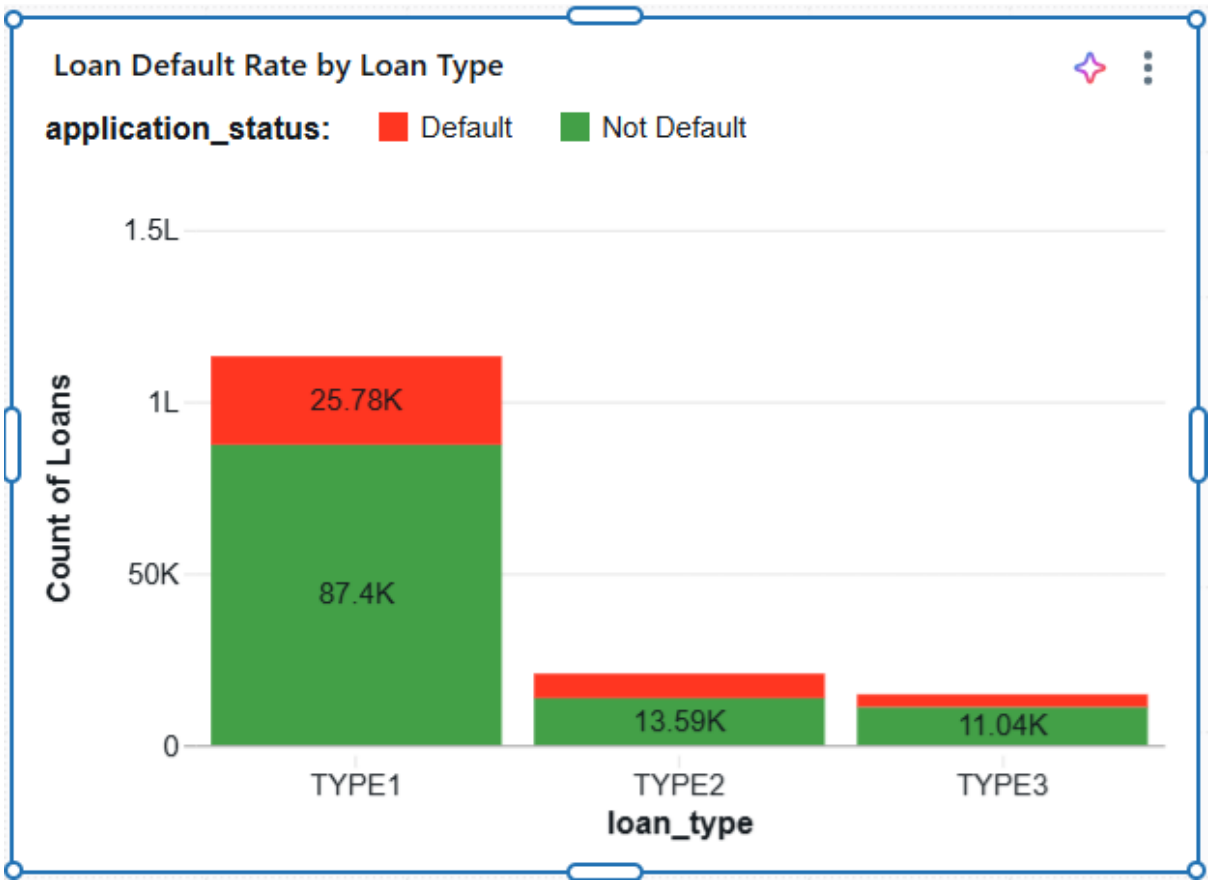
5. **Income vs Loan Risk**

This visualization compares borrower income with loan amount while highlighting risk categories. It helps determine whether higher-income individuals take safer or riskier loans. Patterns such as high loans with low income can indicate potential default risk. The color-coded risk categories make it easier to identify risky segments visually. This insight supports better financial profiling of borrowers. It also helps in setting income-based lending limits.



6. Loan Default Analysis

This chart compares loan defaults across various loan types using application status as a breakdown. It helps in identifying which loan categories are more prone to defaults. By analyzing this, lenders can understand risk patterns associated with different loan products. For example, unsecured loans may show higher default rates than secured ones. This insight is crucial for improving loan approval policies and risk assessment strategies. It also helps in optimizing product offerings.

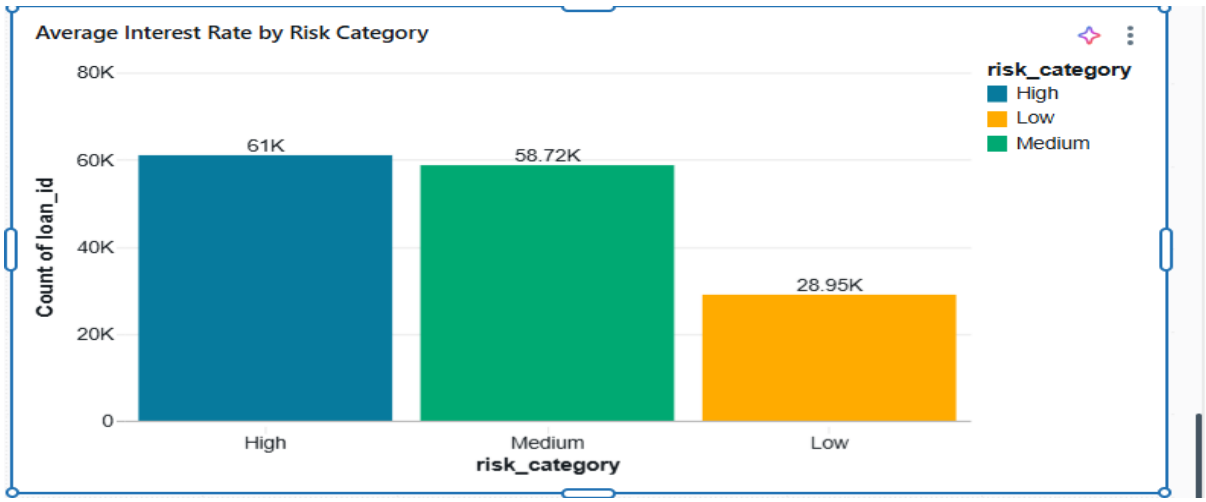


7. **High-Risk Borrower Identification**

This visualization shows the distribution of borrowers across different risk categories such as High, Medium, and Low. It helps in quickly identifying the proportion of high-risk customers in the dataset. By applying filters like risk_score > 0.7, we can focus specifically on critical cases. This insight is useful for financial institutions to prioritize monitoring and mitigation strategies. A higher count in the high-risk category indicates potential default threats. It supports proactive decision-making in credit risk management.

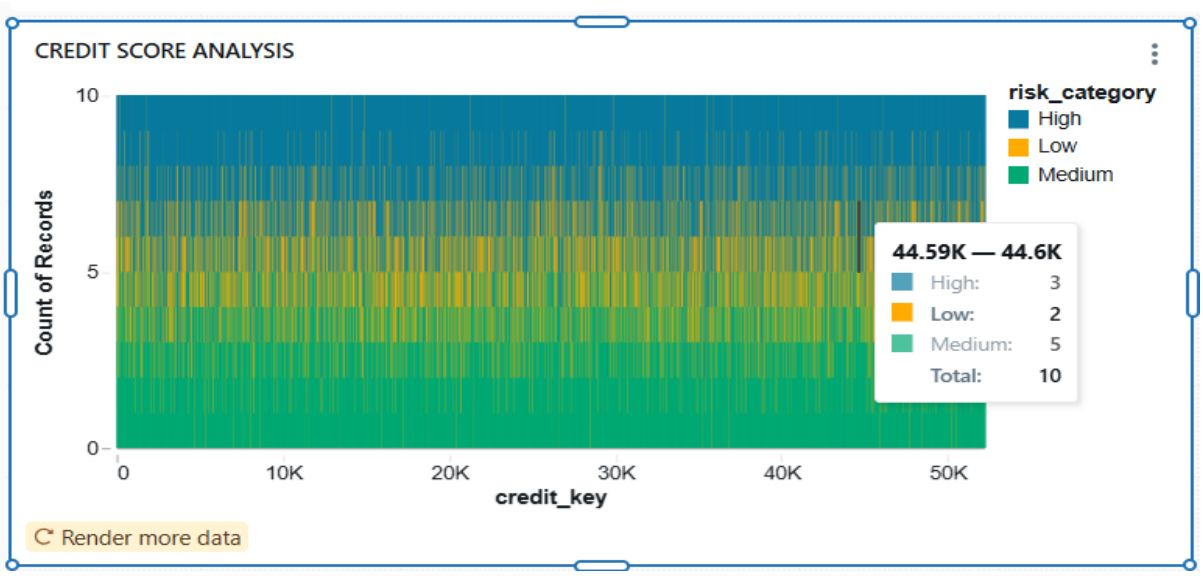
8. Region-wise Average Risk

This visualization calculates the average risk score for each region to provide a summarized risk view. It helps quickly identify regions with higher overall risk exposure. Regions with higher averages may require stricter credit policies. This metric simplifies complex data into an easy-to-understand format. It also supports comparison across regions for better strategic planning. This is a powerful addition for executive-level insights.



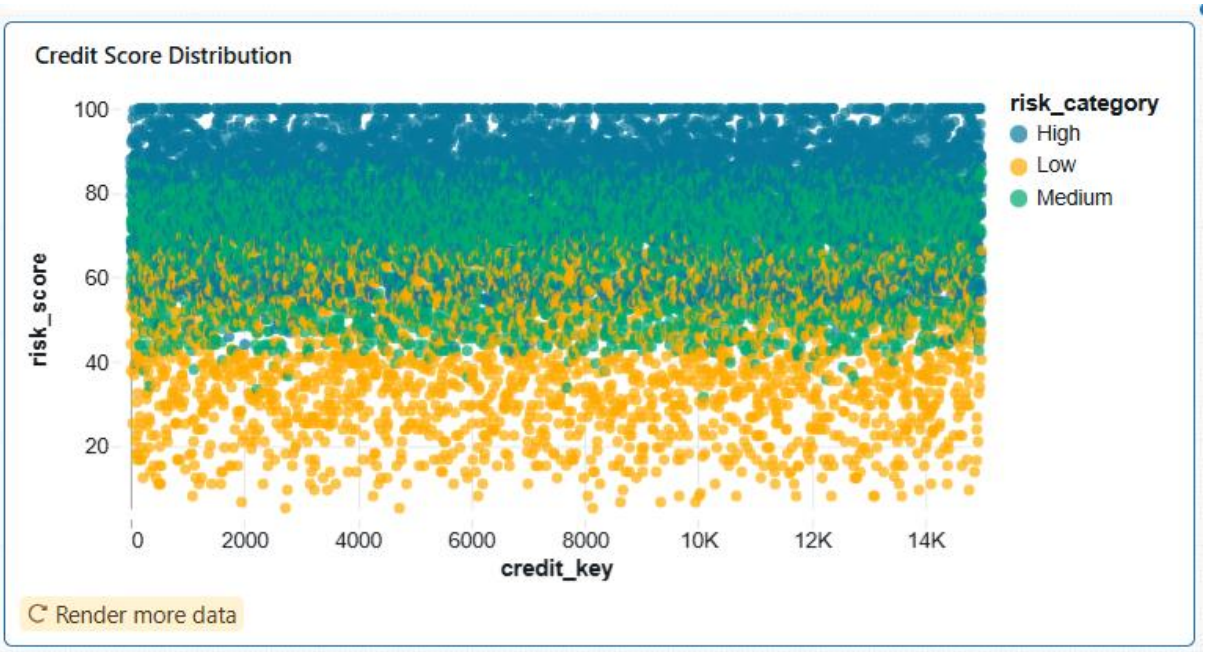
9. Credit Score Analysis

This histogram (or bar chart) groups borrowers based on their credit scores to show distribution patterns. It helps identify whether most customers fall into low, medium, or high credit score ranges. A concentration of low scores may indicate higher portfolio risk. On the other hand, a balanced or high-score distribution suggests better credit quality. Binning helps simplify continuous data into meaningful ranges. This insight is useful for risk segmentation and customer profiling.



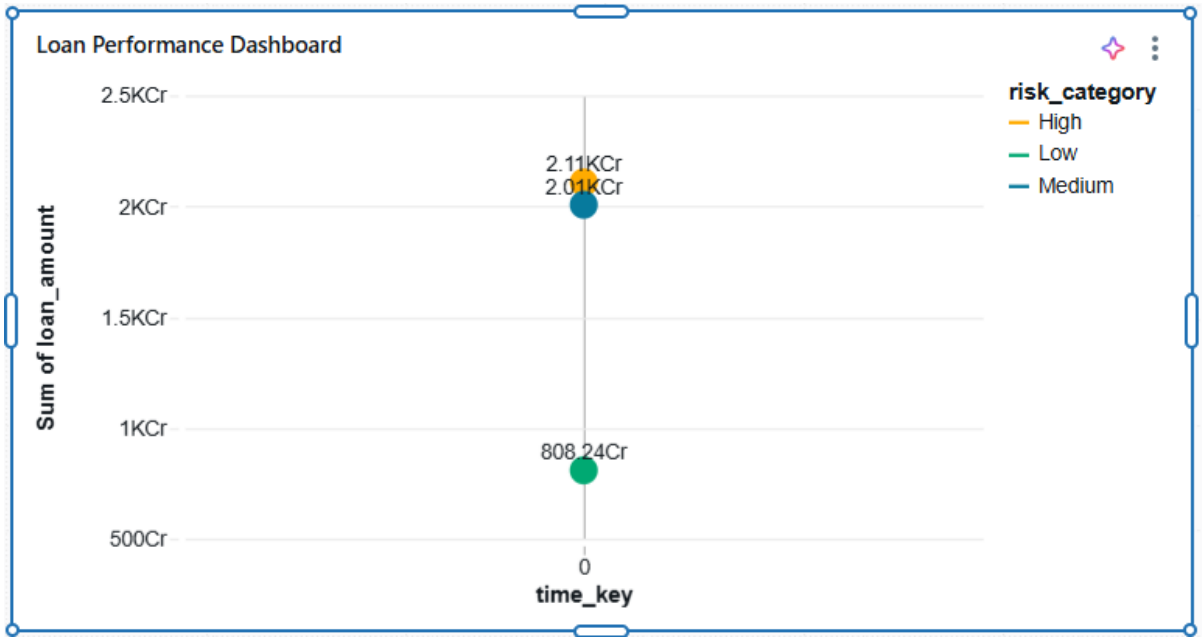
10. Credit Score Distribution

This scatter plot shows the relationship between credit score and risk score, highlighting different risk categories. Typically, borrowers with lower credit scores tend to have higher risk scores. The visualization helps identify trends, clusters, and anomalies in borrower behavior. Color coding by risk category makes it easier to distinguish between high, medium, and low-risk segments. This analysis validates the effectiveness of credit scoring models. It also supports better decision-making in loan approvals.



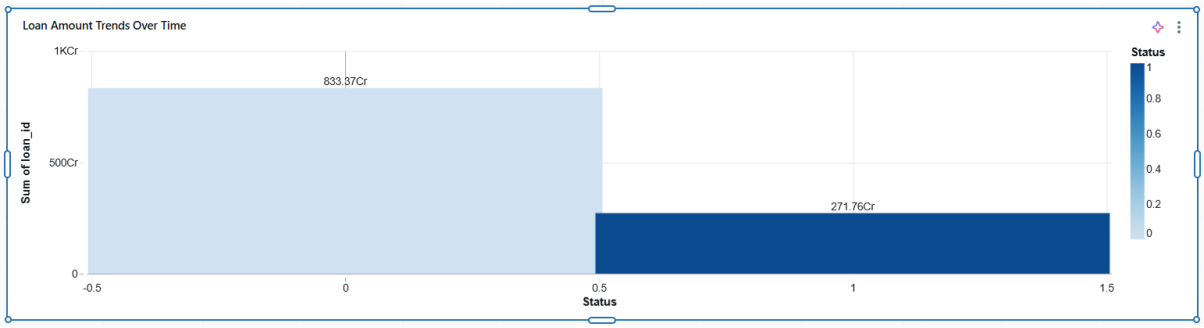
11. Loan Performance Dashboard

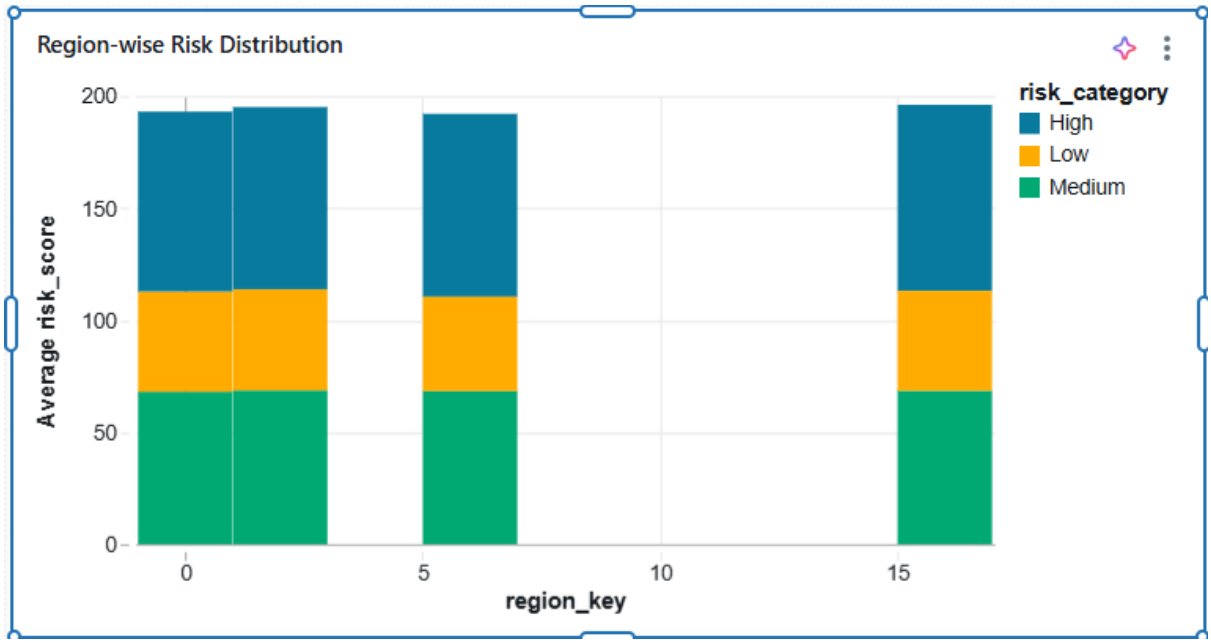
This line chart visualizes the trend of total loan amounts across different time periods using the year dimension. It helps identify growth or decline in lending activity over time. Peaks in the graph may indicate high loan disbursement periods, while drops may signal stricter policies or reduced demand. This insight is useful for understanding business performance and seasonal patterns. It also supports strategic planning and forecasting. Overall, it provides a clear view of lending trends.



12. Loan Amount Trends

his bar chart compares the number of defaulted and non-defaulted loans using application status. It provides a quick understanding of overall loan performance and risk exposure. A higher number of defaults indicates potential issues in credit evaluation or borrower quality. Renaming values to “Default” and “Non-Default” improves clarity for stakeholders. The color distinction makes it easier to interpret patterns visually. This analysis is essential for monitoring portfolio health and improving risk strategies.



13. Region-wise Risk Distribution

This bar chart presents the distribution of loans across regions with a breakdown by risk category. It helps identify which regions have a higher concentration of high-risk borrowers. Stacked bars make it easy to compare risk composition within each region. This insight is useful for region-specific risk management and policy adjustments. It also helps financial institutions focus on high-risk geographic areas. Overall, it supports localized decision-making.